

Chapter 1

Black-Scholes Model

1.1 Model Setup: Stock and Bond Dynamics

Call option price formula is shown by

$$C = S_0 \mathbb{N}(d_1) - Ke^{-rT} \mathbb{N}(d_2)$$

Put option price can be calculated by Put-Call Parity

$$P + S_0 = C + Ke^{-rT}$$

Assuming we have a portfolio with underlying stock S and risk-free asset B , differential equations are shown by

$$dB_t = rB_t dt \quad (1.1)$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1.2)$$

where $dW_t = \varepsilon \sqrt{dt}$; $\varepsilon \sim \mathcal{N}(0,1)$, which is a Ito process. μ is drift (deterministic) and dW_t is a diffusion (uncertainty).

Solving S_t and B_t from differential equations

B_t and S_t can be expressed as

$$B_t = B_0 e^{rt} \quad (1.3)$$

Proof:

$$\begin{aligned} \int \frac{dB_t}{B_t} &= \int r dt \\ \int_0^t \ln B_t &= \int_0^t r dt \\ \ln(B_t/B_0) &= r * t \\ B_t &= B_0 e^{rt} \end{aligned}$$

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$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right] \quad (1.4)$$

Proof: Let $Y_t = \ln S_t$

$$\begin{aligned} dY_t &= \frac{\partial Y_t}{\partial t} dt + \frac{\partial Y_t}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 Y_t}{\partial S_t^2} dS_t^2 \\ &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t \end{aligned}$$

we get

$$\begin{aligned} Y_t &= Y_0 + \int_0^t (\mu - \frac{1}{2}\sigma^2)ds + \int_0^t \sigma dW_s \\ &= Y_0 + (\mu - \frac{1}{2})t + \sigma W_t \end{aligned}$$

Given $Y_t = \ln S_t$ Finally, we have to show

$$\begin{aligned} \ln S_t &= \ln S_0 + (\mu - \frac{1}{2})t + \sigma W_t \\ \implies S_t &= S_0 \exp \left[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t \right] \end{aligned}$$

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1.2 Risk-Neutral Martingale Pricing Derivation

Proposition 1.2.1

In the Black-Scholes model, the price process Π_t for every traded asset, be it the underlying or derivative asset, has the property that the normalised price process

$$Z_t = \frac{\Pi_t}{B_t}$$

is a martingale under the $\mathbb{Q}$ measure.

In this case,

$$Z_t = \frac{S_t}{B_t}$$

Its differential equation can be written as

$$dZ_t = d\left(\frac{S_t}{B_t}\right)$$

$d\left(\frac{1}{B_t}\right)$ can be calculated by Ito Lamma

$$\begin{aligned} d\left(\frac{1}{B_t}\right) &= 0dt + \left(-\frac{1}{B_t^2}\right)dB_t + \frac{1}{2} \cdot \left(-\frac{2}{B_t^3}\right)(dB_t)^2 \\ &= -\frac{1}{B_t^2} \cdot rB_t dt - 0 \\ &= -\frac{1}{B_t} r dt \end{aligned}$$

Using product rule, then we have

$$\begin{aligned} d\left(\frac{S_t}{B_t}\right) &= \frac{1}{B_t} \cdot dS_t + S_t \cdot d\left(\frac{1}{B_t}\right) + dS_t \cdot d\left(\frac{1}{B_t}\right) \\ &= \frac{1}{B_t} \cdot (\mu S_t dt + \sigma S_t dW_t) + S_t \cdot \left(-\frac{r}{B_t}\right) dt + 0 \\ &= \frac{S_t}{B_t} \mu dt + \frac{S_t}{B_t} \sigma dW_t - \frac{S_t}{B_t} r dt \\ &= \frac{S_t}{B_t} (\mu - r) dt + \frac{S_t}{B_t} \sigma dW_t \end{aligned}$$

Namely under $\mathbb{P}$ measure

$$dZ_t = Z_t(\mu - r)dt + Z_t\sigma dW_t$$

Transfer $\mathbb{P}$ to $\mathbb{Q}$ measure

Theorem 1.2.1 Novikov Condition

θ satisfies the Novikov Condition if $\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \theta_s^2 ds \right) \right] < \infty$

If θ satisfies Novikov condition, the process $\mathcal{M}^\theta$ defined as

$$\mathcal{M}_t^\theta = \exp \left(\int_0^t \theta_s dX_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right), t \in [0, T]$$

is a martingale.

Theorem 1.2.2 Girsanovs Theorem

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(\int_0^t \theta_s dX_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right)$$

In this case, the process $X^\mathbb{Q}$ is defined as

$$X_t^\mathbb{Q} = X_t^\mathbb{P} - \int_0^t \theta_s ds, t \in [0, T]$$

is a standard Brownian Motion.

From Girsanovs Theorem, we have

$$W_t^\mathbb{Q} = W_t^\mathbb{P} + \int_0^t \theta_s ds$$

and

$$dW_t^\mathbb{Q} = dW_t + \theta_t dt$$

$$\begin{aligned} dZ_t &= (\mu - r)Z_t dt + \sigma Z_t dW_t^\mathbb{P} \\ &= (\mu - r)Z_t dt + \sigma Z_t (\theta_t dt + dW_t^\mathbb{Q}) \\ &= (\mu - r + \sigma \theta_t)Z_t dt + \sigma Z_t dW_t^\mathbb{Q} \end{aligned}$$

dZ_t is a martingale under $\mathbb{Q}$ dynamics Therefore

$$\mu - r + \sigma \theta_t = 0$$

θ_t can be expressed as

$$\theta_t = \frac{\mu - r}{\sigma}$$

And θ_t is the Sharpe Ratio. μ is the return of underlying stock S and r is the risk-free rate.

Under $\mathbb{Q}$ measure

$$dZ_t = \sigma Z_t dW_t^\mathbb{Q}$$

Portfolio Replication

Return back to our portfolio. Φ_t^S and Φ_t^B are the weights of the stock and risk-free asset respectively.

The value of the portfolio V_t and its differential equation are shown as

$$V_t = \Phi_t^S S_t + \Phi_t^B B_t$$

$$dV_t = \Phi_t^S dS_t + \Phi_t^B dB_t$$

Checking if V_t is a martingale

$$\begin{aligned}
d\left(\frac{V_t}{B_t}\right) &= dV_t * \frac{1}{B_t} + V_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^S dS_t + \Phi_t^B dB_t * \frac{1}{B_t} + V_t + V_t = \Phi_t^S S_t + \Phi_t^B B_t * d\left(\frac{1}{B_t}\right) \\
&= \Phi_t^S dS_t * \left(\frac{1}{B_t}\right) + \Phi_t^B dB_t * \left(\frac{1}{B_t}\right) + \Phi_t^S S_t * \left(-\frac{1}{B_t}\right) r dt + \Phi_t^B B_t * \left(-\frac{1}{B_t}\right) r dt \\
&= \Phi_t^S \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t \right] + \Phi_t^B \left[\left(\frac{1}{B_t}\right) * r B_t dt - r B_t * \left(\frac{1}{B_t}\right) dt \right] \\
&= \Phi_t^S \left[-r S_t * \left(\frac{1}{B_t}\right) dt + \left(\frac{1}{B_t}\right) dS_t \right] \\
&= \Phi_t^S \left[S_t * d\left(\frac{1}{B_t}\right) + \left(\frac{1}{B_t}\right) * dS_t \right] \\
&= \Phi_t^S * d\left(\frac{S_t}{B_t}\right) \\
&= \Phi_t^S dZ_t
\end{aligned}$$

Under $\mathbf{Q}$ measure

$$d\left(\frac{V_t}{B_t}\right) = \Phi_t^S dZ_t = \Phi_t^S * \sigma Z_t dW_t^{\mathbf{Q}}$$

Let

$$G_t = \frac{V_t}{B_t}$$

Then G_t is a martingale under $\mathbf{Q}$ measure

Remark the definition of martingale $\forall s < t, \mathbb{E}[X_t | \mathcal{F}_s] = X_s$, X_t is a Martingale.

Therefore

$$\mathbb{E}^{\mathbf{Q}}[G_T | \mathcal{F}_t] = G_t$$

$\chi(t, S_t)$ is the option value. At time T , the option value is $\chi(T, S_T)$, S_T is the underlying price at time T .

$$\begin{aligned}
V_t &= G_t * B_t \\
&= e^{-rt} \mathbb{E}^{\mathbf{Q}}[G_T | \mathcal{F}_t] \\
&= e^{-rt} \mathbb{E}^{\mathbf{Q}}[V_T * e^{rT} | \mathcal{F}_t] \\
&= e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[V_T | \mathcal{F}_t] \\
&= e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\} | \mathcal{F}_t]
\end{aligned}$$

Therefore

$$V_t = e^{-r(T-t)} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\} | \mathcal{F}_t]$$

Pricing Call Option

We want to price the value of a call option at $t = 0$

$$V_0 = e^{-rT} \mathbb{E}^{\mathbf{Q}}[\max\{S_T - K, 0\}]$$

Let's introduce an indicator function

$$V_0 = e^{-rT} \mathbb{E}^{\mathbf{Q}}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}]$$

Under $\mathbf{P}$ measure

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbf{P}}$$

Under $\mathbb{Q}$ measure

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t^{\mathbb{Q}} \\ S_t &= S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma \varepsilon \sqrt{t}} \end{aligned}$$

and

$$\begin{aligned} S_T &= S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \varepsilon \sqrt{T}} \\ V_0 &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}] \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T \geq K\}}] - e^{-rT} \mathbb{E}^{\mathbb{Q}}[K \mathbf{1}_{\{S_T \geq K\}}] \end{aligned}$$

Lemma 1.2.1 Change of Measure Identity

$$\mathbb{E} \left[e^{m + \lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}} \right] = e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a)$$

where $\varepsilon \sim \mathcal{N}(0, 1)$, m, λ, a are constant.

Proof:

$$\begin{aligned} \mathbb{E} \left[e^{m + \lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}} \right] &= \int_{-\infty}^{\infty} e^{m + \lambda x} \mathbf{1}_{\{x \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \int_a^{\infty} e^{m + \lambda x} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \int_a^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\lambda)^2}{2}} e^{\frac{\lambda^2}{2} + m} dx \\ &= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} e^{\frac{\lambda^2}{2} + m} dy \\ &= e^{\frac{\lambda^2}{2} + m} [\mathcal{N}(\infty) - \mathcal{N}(a - \lambda)] \\ &= e^{\frac{\lambda^2}{2} + m} [1 - \mathcal{N}(a - \lambda)] \\ &= e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a) \end{aligned}$$

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$$\begin{aligned} e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T \geq K\}}] &= e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma \varepsilon \sqrt{T} \right] \mathbf{1}_{\{S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma \varepsilon \sqrt{T}] \geq K\}} \right] \\ &= e^{-rT} S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) T \right] \mathbb{E} \left[\exp^{\sigma \varepsilon \sqrt{T}} \mathbf{1}_{\left\{ \varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right\}} \right] \\ &= S_0 \exp \left(-\frac{1}{2}\sigma^2 T \right) \mathbb{E}^{\mathbb{Q}} \left[\exp^{0 + \sigma \sqrt{T} \varepsilon} \mathbf{1}_{\left\{ \varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right\}} \right] \\ &= S_0 \exp \left(-\frac{1}{2}\sigma^2 T \right) * \exp \left(\frac{\sigma^2 T}{2} + 0 \right) * \mathcal{N} \left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 * \mathcal{N} \left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 \mathcal{N} \left(\frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} \right) \\ &= S_0 \mathcal{N}(d_1) \end{aligned}$$

$$\begin{aligned}
e^{-rT} \mathbb{E} [K \mathbf{1}_{\{S_T \geq K\}}] &= K e^{-rT} \mathbb{E} [\mathbf{1}_{\{S_T \geq K\}}] \\
&= K e^{-rT} [1 * Q(S_T \geq K) + 0 * Q(S_T < K)] \\
&= K e^{-rT} * Q(S_T \geq K) \\
&= K e^{-rT} Q\left(\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}\left(\frac{\ln(\frac{S_0}{K}) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}(d_2)
\end{aligned}$$

Therefore, the call option price

$$V_0 = S_0 \mathbf{N}(d_1) - K e^{-rT} \mathbf{N}(d_2)$$

1.3 Delta-Hedging PDE Derivation

Considering we have an option and we want to eliminate the risk by buying some shares of stock.

$$\Pi = V - \Delta S$$

Its differential equation is

$$d\Pi = dV - \Delta dS$$

Remark dS and dV

$$\begin{aligned}
dS_t &= \mu S_t dt + \sigma S_t dW_t \\
dV_t &= \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt + \frac{\partial V}{\partial S} dS
\end{aligned}$$

Plugging dV_t to $d\Pi$, we have

$$d\Pi_t = \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt + \left(\frac{\partial V}{\partial S} - \Delta \right) dS$$

Since

$$\Delta = \frac{\partial V}{\partial S}$$

Then

$$d\Pi_t = \left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt$$

Since the hedged portfolio is locally riskless, by no-arbitrage it must earn the risk-free rate. We have

$$d\Pi_t = \Pi r dt$$

Therefore

$$\left\{ \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right\} dt = r \left(V - S \frac{\partial V}{\partial S} \right) dt$$

Removing right side two terms to left side, we get

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$

Theorem 1.3.1 Feynman-Kac

Assume that F is a solution to the boundary value problem:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$
$$F(T, x) = \Phi(S)$$

Then F has the representation

$$F(t, x) = e^{-r(T-t)} \mathbb{E}[\Phi(S(T))]$$

The call option price is

$$V = e^{-rT} \mathbb{E}[(S_T - K)^+]$$

1.4 Self-Financing Replication Derivation

We set the χ is a option, and $G(S_t)$ is the value at the expiry date.
 V_t is the value of the option at the expiry date.

$$V_t = V_0 + \int_0^t \Phi_u^S dS_u + \Phi_u^B dB_u \quad (1.5)$$

Note:-

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

To derivative Black-Scholes, we have to solutions that kills dW_t or dt .

1. kill dW_t and remain dt , hedge risk (uncertainty) long call and short Δ stock (risk management) $\rightarrow PDE$
2. kill dt and remain dW_t , we get $f(t, S_t)dW_t$, which is a martingale

Set

$$S_t^* = S_t e^{-rt}$$

$$\begin{array}{ccc} | & & | \\ S_t^* & & S_t \end{array} \quad \begin{array}{c} | \\ S_T \end{array}$$

$$d(X_t, Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$$

$$de^{-rt} = -re^{-rt} dt$$

$$dt^2 = 0$$

$$dt dW_t = 0$$

$$dW_t^2 = dt$$

we have

$$\begin{aligned} dS_t^* &= d(S_t e^{-rt}) \\ &= S_t de^{-rt} + e^{-rt} dS_t + dS_t de^{-rt} \\ &= S_t(-e^{-rt} dt) + e^{-rt}(\mu S_t dt + \sigma S_t dW_t) \\ &= (\mu - r)S_t^* dt + \sigma S_t^* dW_t \end{aligned}$$

In general, $\mu - r > 0$ under $\mathbf{P}$ measurement. We can find $\mathbf{Q}$ st. $\{S_t^*, 0 \leq t \leq T\}$ is a $\mathbf{Q}$ -Martingale, where $\mu - r = 0$.
 We need to prove $dS_t^* = \sigma S_t^* dW_t$

The θ_s is unknown, we need to solve it.

$$\begin{aligned} dS_t^* &= (\mu - r)S_t^* dt + \sigma S_t^* dW_t \\ &= (\mu - r)S_t^* dt + \sigma S_t^* (dW_t^{\mathbf{Q}} - \theta_t dt) \\ &= \mu S_t^* dt - r S_t^* dt + \sigma S_t^* dW_t^{\mathbf{Q}} - \sigma S_t^* \theta_t dt \\ &= (\mu - r - \sigma \theta_t)S_t^* dt + \sigma S_t^* dW_t^{\mathbf{Q}} \end{aligned}$$

Let $\mu - r - \theta_t = 0$ to kill drift (dt), under $\theta_t = \frac{(\mu - r)}{\sigma}$ (Sharpe Ratio)

Finally, we get

$$dS_t^* = \sigma S_t^* dW_t^{\mathbf{Q}}$$

and we conclude that S_t^* is a $\mathbf{Q}$ -Martingale.

Definition 1.4.1: Martingale

$\forall s < t \implies \mathbb{E}[X_t | \mathcal{F}_s] = X_s$, X_t is a Martingale.

$\chi(t, S_t)$ is the option value and V_t is the value of the portfolio replicated by risk-free asset and stocks. So, $\chi(t, S_t) = V_t$. We set V_t^* is the discounted value of V_t , where $V_t^* = V_t e^{-rt}$.

$$\begin{aligned}
dV_t^* &= d(V_t e^{-rt}) \\
&= V_t de^{-rt} + e^{-rt} dV_t + dV_t de^{-rt} \\
&= (\Phi_t^S S_t + \Phi_t^B B_t) r e^{-rt} dt + (\Phi_t^S S_t + \Phi_t^B B_t) e^{-rt} d\ln V_t \\
&= \Phi_t^S (-r S_t + e^{-rt} dt + e^{-rt} dS_t) \\
&= \Phi_t^S (S_t de^{-rt} + r^{-rt} dS_t) \\
&= \Phi_t^S d(S_t e^{-rt}) \\
&= \Phi_t^S dS_t^*
\end{aligned}$$

$$dV_t^* = \Phi_t^S dS_t^*$$

Given $dS_t^* = \sigma S_t^* dW_t^Q$
We get

$$dV_t^* = \Phi_t^S \sigma S_t^* dW_t^Q$$

$$\mathbb{E}[V_T | \mathcal{F}_t] = V_t^*$$

$$V_T^* = V_T e^{-rT} = \chi(T, S_T) e^{-rT} = G(S_T) e^{-rT}$$

$$\mathbb{E}[V_T^* | \mathcal{F}_t] = \mathbb{E}[G(S_T) e^{-rT} | \mathcal{F}_t]$$

Claim 1.4.1 Call Option Pricing

$$C = e^{-rT} \mathbb{E}[(S_T - K) \mathbf{1}_{\{S_T \geq K\}}]$$

Lemma 1.4.1

$$\mathbb{E} [e^{m+\lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}}] = e^{\frac{\lambda^2}{2} + m} \mathbf{N}(\lambda - a)$$

where $\varepsilon \sim \mathbf{N}(0, 1)$, m, λ, a are constant.

Proof:

$$\begin{aligned}
\mathbb{E} [e^{m+\lambda \varepsilon} \mathbf{1}_{\{\varepsilon \geq a\}}] &= \int_{-\infty}^{\infty} e^{m+\lambda x} \mathbf{1}_{\{x \geq a\}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&= \int_a^{\infty} e^{m+\lambda x} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\
&= \int_a^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\lambda)^2}{2}} e^{\frac{\lambda^2}{2} + m} dx \\
&= \int_{a-\lambda}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} e^{\frac{\lambda^2}{2} + m} dy \\
&= e^{\frac{\lambda^2}{2} + m} [\mathcal{N}(\infty) - \mathcal{N}(a - \lambda)] \\
&= e^{\frac{\lambda^2}{2} + m} [1 - \mathcal{N}(a - \lambda)] \\
&= e^{\frac{\lambda^2}{2} + m} \mathcal{N}(\lambda - a)
\end{aligned}$$

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Proof:

$$\begin{aligned}
e^{-rT} \mathbb{E}^Q[S_T \mathbf{1}_{\{S_T \geq K\}}] &= e^{-rT} \mathbb{E}^Q \left[S_0 \exp\left[\left(r - \frac{1}{2}\right)T + \sigma \varepsilon \sqrt{T}\right] \mathbf{1}_{\{S_0 \exp\left[\left(r - \frac{1}{2}\right)T + \sigma \varepsilon \sqrt{T}\right] \geq K\}} \right] \\
&= e^{-rT} S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)T\right] \mathbb{E} \left[\exp^{\sigma \varepsilon \sqrt{T}} \mathbf{1}_{\left\{\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right\}} \right] \\
&= S_0 \exp\left(-\frac{1}{2}\sigma^2 T\right) \mathbb{E}^Q \left[\exp^{0 + \sigma \sqrt{T} \varepsilon} \mathbf{1}_{\left\{\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right\}} \right] \\
&= S_0 \exp\left(-\frac{1}{2}\sigma^2 T\right) * \exp\left(\frac{\sigma^2 T}{2} + 0\right) * \mathbf{N}\left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 * \mathbf{N}\left(\sigma \sqrt{T} - \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 \mathbf{N}\left(\frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= S_0 \mathbf{N}(d_1)
\end{aligned}$$

$$\begin{aligned}
e^{-rT} \mathbb{E}[K \mathbf{1}_{\{S_T \geq K\}}] &= K e^{-rT} \mathbb{E}[\mathbf{1}_{\{S_T \geq K\}}] \\
&= K e^{-rT} [1 * Q(S_T \geq K) + 0 * Q(S_T < K)] \\
&= K e^{-rT} * Q(S_T \geq K) \\
&= K e^{-rT} Q\left(\varepsilon \geq \frac{\ln(\frac{K}{S_0}) - (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}\left(\frac{\ln(\frac{S_0}{K}) + (r - \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}}\right) \\
&= K e^{-rT} \mathbf{N}(d_2)
\end{aligned}$$

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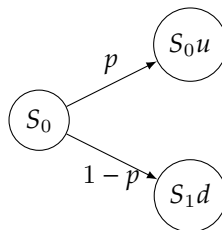
Claim 1.4.2 Call Option Pricing

$$C = S_0 \mathbf{N}(d_1) - K e^{-rT} \mathbf{N}(d_2)$$

1.5 Bridging the Gap: Binomial Tree Delta-Hedging and its Convergence to Black-Scholes

Discrete Binomial Tree

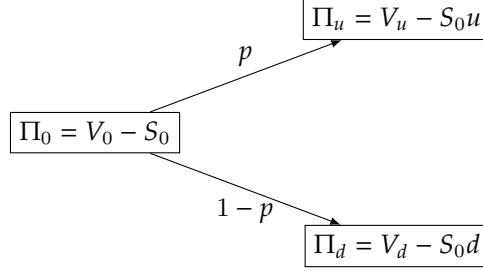
Consider a single period binomial tree



We have a portfolio with a option and stock, V and S are the values respectively. Π is the portfolio value. We want to use delta hedging to elimiate the risk, which is shown by

$$\Pi = V - \Delta S$$

where Δ is the number of the shares.



We want to pick a Δ to make the equation holding

$$\Pi_u = \Pi_d, \quad V_u - S_0 u = V_d - S_0 d$$

The Δ is solved by

$$\Delta = \frac{V_u - V_d}{S_0(u - d)}$$

Assuming maturity is δt and the discounted factor

$$D = \frac{1}{B} = e^{-r\delta t}$$

Under assumption of arbitrage free,

$$\Pi_0 = e^{-r\delta t}[V_u - \Delta S_u] = e^{-r\delta t}[V_d - \Delta S_d]$$

Namely

$$V_0 - \Delta S_0 = e^{-r\delta t}[V_u - \Delta S_0 u] = e^{-r\delta t}[V_d - \Delta S_0 d]$$

V_0 is solved as

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

In risk neutral world, investors must earn the risk-free rate on average. We have

$$\begin{aligned} S_0 e^{r\delta t} &= \mathbb{E}[S_{\delta t}] \\ &= q * S_0 u + (1 - q) S_0 d \end{aligned}$$

The probability of moving up under $\mathbf{Q}$ is

$$q = \frac{e^{r\delta t} - d}{u - d}$$

The option price V value at $t = 0$ is

$$V_0 = e^{-r\delta t}[q * V_u + (1 - q) * V_d]$$

While the replication argument is based on hedging, we can simplify the formula by introducing a synthetic probability q . We define the "Risk-Neutral World" as a measure $\mathbf{Q}$ where the expected return of the stock is the risk-free rate:

$$S_0 e^{r\delta t} = \mathbb{E}^{\mathbf{Q}}[S_{\delta t}] = q S_0 u + (1 - q) S_0 d$$

Solving for q (the risk-neutral probability of an upward move):

$$q = \frac{e^{r\delta t} - d}{u - d}$$

Under this measure $\mathbf{Q}$, the option price becomes a simple discounted expectation:

$$V_0 = e^{-r\delta t} \mathbb{E}^{\mathbf{Q}}[V_{\delta t}] = e^{-r\delta t}[q V_u + (1 - q) V_d]$$

Therefore, the option price at $t = 0$ can be written as

$$V_0 = \mathbb{E}^Q[e^{-r\delta t} V_T] = e^{-r\delta t} [q * V_u + (1 - q)V_d]$$

Since Cox-Ross-Rubinstein, we have

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = \frac{1}{u}, \quad D = e^{-r\delta t}$$

$$V_0 = \frac{V_u(1 - de^{-r\delta t}) - V_d(ue^{-r\delta t} - 1)}{u - d}$$

Multiply by $(u - d) * e^{r\delta t}$ both sides

$$(u - d)V_0 * e^{r\delta t} = e^{r\delta t}(V_u - V_d) + (uV_d - dV_u)$$

Using Taylor Expansion, we have

$$u = e^{\sigma\sqrt{\delta t}} \approx 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$$

$$d = \frac{1}{u} \approx 1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$$

$$e^{r\delta t} \approx 1 + r\delta t$$

Continuous Binomial Tree

We follow the CRR (Cox-Ross-Rubinstein) Convergence path.

Step 1: The Multi-period Binomial Formula In a binomial tree with n steps, the option price V_0 is the discounted expected value of all possible payoff paths. Let k be the number of "up" moves:

$$V_0 = e^{-rT} \sum_{k=0}^n \binom{n}{k} q^k (1 - q)^{n-k} \max(S_0 u^k d^{n-k} - K, 0)$$

Let a be the minimum number of up moves required for the option to finish in-the-money ($S_T \geq K$). We split the sum into two parts:

$$V_0 = S_0 \left[\sum_{k=a}^n \binom{n}{k} q^k (1 - q)^{n-k} \frac{u^k d^{n-k}}{e^{rT}} \right] - K e^{-rT} \left[\sum_{k=a}^n \binom{n}{k} q^k (1 - q)^{n-k} \right]$$

Step 2: Parameter Setup (CRR) and Taylor Expansion To ensure the discrete model matches the Geometric Brownian Motion (GBM) as $\delta t \rightarrow 0$, we use:

$$u = e^{\sigma\sqrt{\delta t}}, \quad d = e^{-\sigma\sqrt{\delta t}}, \quad q = \frac{e^{r\delta t} - d}{u - d}$$

Using Taylor Expansion $e^x \approx 1 + x + \frac{1}{2}x^2$: $u \approx 1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$, $d \approx 1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t$, $e^{r\delta t} \approx 1 + r\delta t$. Substituting these into q :

$$q \approx \frac{(1 + r\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}{(1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)} \approx \frac{\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t}{2\sigma\sqrt{\delta t}} = \frac{1}{2} + \frac{1}{2\sigma} \left(r - \frac{1}{2}\sigma^2 \right) \sqrt{\delta t}$$

Step 3: Application of Central Limit Theorem (CLT) As $n \rightarrow \infty$, the binomial distribution $B(n, q)$ converges to a Normal Distribution $N(\mu, \sigma^2)$. The probability of the option being in-the-money (the second bracket in Step 1) is:

$$P(k \geq a) = P\left(\frac{k - nq}{\sqrt{nq(1 - q)}} \geq \frac{a - nq}{\sqrt{nq(1 - q)}} \right)$$

Using the CLT, this probability converges to the cumulative normal distribution:

$$\lim_{n \rightarrow \infty} P(k \geq a) = N(d_2)$$

Where $d_2 = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$. Step 4: The Change of Numeraire for $N(d_1)$ The first bracket in Step 1 involves an extra term $u^k d^{n-k}$. We define a new probability measure q' (often called the stock measure):

$$q' = q \frac{u}{e^{r\delta t}}$$

By substituting q' into the summation, we get a similar normal convergence:

$$\lim_{n \rightarrow \infty} \sum_{k=a}^n \binom{n}{k} (q')^k (1 - q')^{n-k} = \mathbf{N}(d_1)$$

Where $d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$. Final Convergence Result Combining the limits from Step 3 and Step 4 back into the Step 1 formula, we arrive at the Black-Scholes Call Price:

$$C = S_0 \mathbf{N}(d_1) - Ke^{-rT} \mathbf{N}(d_2)$$

1.6 Comparison of the Three Perspectives

The Black-Scholes formula can be derived from several different perspectives. Although these derivations look different, they are not completely independent. They are different mathematical representations of the same no-arbitrage principle in a complete market.

The risk-neutral martingale approach derives the option price by changing from the physical probability measure $\mathbf{P}$ to the risk-neutral measure $\mathbf{Q}$. Under $\mathbf{Q}$, the discounted stock price is a martingale. Therefore, the derivative price can be written as the discounted expected payoff:

$$V_t = e^{-r(T-t)} \mathbf{E}^{\mathbf{Q}}[V_T | \mathcal{F}_t].$$

This approach gives the most compact pricing formula and is especially useful for modern asset pricing theory. The delta-hedging approach derives the Black-Scholes partial differential equation by constructing a locally riskless portfolio. By choosing

$$\Delta = \frac{\partial V}{\partial S},$$

the stochastic term driven by dW_t is eliminated. The remaining portfolio is locally riskless and, by the no-arbitrage principle, must earn the risk-free rate. This leads to the Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

The self-financing replication approach shows that the option payoff can be replicated by dynamically trading the stock and the risk-free bond. Because the replicating portfolio has the same terminal payoff as the option, the no-arbitrage principle implies that the option price must equal the cost of the replicating strategy. This approach gives the strongest economic interpretation of option pricing: the option value is the cost of replication.

These three perspectives are closely connected. Delta hedging constructs the local riskless position, self-financing replication extends this idea into a dynamic trading strategy, and martingale pricing expresses the same no-arbitrage price as a discounted expectation under the risk-neutral measure.

Other derivations, such as the Feynman-Kac representation, Girsanov theorem, heat-equation transformation, lognormal integration, change of numeraire, and the binomial tree limit, can be viewed as refinements or extensions of these core ideas. For example, Girsanov theorem explains how the physical measure $\mathbf{P}$ is transformed into the risk-neutral measure $\mathbf{Q}$; Feynman-Kac connects the PDE solution to the expectation formula; and the binomial tree shows how discrete-time hedging converges to the continuous-time Black-Scholes model.

Therefore, the key message is that Black-Scholes pricing is not based on forecasting the real-world expected return of the stock. Instead, it is based on no-arbitrage, dynamic hedging, and pricing under an equivalent risk-neutral measure.

Comment

A useful way to understand the Black–Scholes model is that it removes the need to estimate the real-world drift μ of the stock. In the delta-hedging derivation, μ disappears because the stochastic risk is hedged away. In the martingale derivation, μ is replaced by the risk-free rate r after changing from $\mathbf{P}$ to $\mathbf{Q}$. In the replication derivation, the option price is determined by the cost of constructing a self-financing portfolio with the same payoff.

Therefore, the Black–Scholes price is not a prediction of the future stock price. It is a no-arbitrage price implied by continuous trading, complete markets, and the ability to dynamically replicate the option payoff.